

Mathematics-assisted directed evolution and protein engineering

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Climate change is resulting in more frequent and intense heatwaves, droughts, and extreme weather events, rising sea levels, loss of biodiversity, and more severe natural disasters. These changes are having significant impacts on human societies, economies, food security, and the environment. Directed evolution, a powerful molecular biology technique that can engineer organisms for environmental remediation (see Figure 1), may be able to mitigate some of the effects of climate change. Directed evolution has also been applied in protein engineering to design enzymes for industrial processes, develop novel therapeutics, and control harmful pathogens [1]. This approach has revolutionized the field of protein engineering, allowing researchers to create proteins with tailored properties and functions that would be difficult or impossible to achieve using traditional methods. However, directed evolution faces significant challenges due to the astronomically large mutational space involved, e.g., 20^{200} for a relatively small protein with 200 amino acids. Additionally, organisms typically have numerous different proteins, making most directed evolution tasks intractable.

Recently, there has been significant interest in artificial intelligence (AI)-assisted directed evolution (AIDE) in the research community [2]. AI has proven to be an effective tool for exploring the protein fitness landscape and identifying optimal evolutionary paths. When combined with experimental validation, AIDE offers an alternative to rational protein design, potentially leading to more effective antibodies and enzymes, as well as flood-, drought-, and disease-resistant plants. Furthermore, AIDE has contributed to a deeper understanding of evolutionary principles. Although AIDE primarily concentrates on protein sequence data, structure data offers a wealth of biophysical information that cannot be extracted from sequences, such as crucial structure-function relationships. The main challenge in utilizing structure data stems from its structural complexity, high dimensionality, nonlinearity, and the multiscale and multi-physical interactions inherent in biological systems.

In recent years, the impact of topological data analysis (TDA) on science and engineering has grown exponentially. The main tool of TDA, persistent homology (PH) [3, 4], bridges the gap between complex geometry and abstract topology through filtration. PH has been incredibly successful in handling intricately complex, high-dimensional, nonlinear, and multiscale data, including those from computer-aided drug design¹. However, it has limitations, including its inability to handle heterogeneous information (i.e., different types of atoms in proteins), its qualitative nature (e.g., ignoring the difference between a 5-member ring and a 6-member ring), its lack of description of non-topological changes (i.e., homotopic shape evolution),

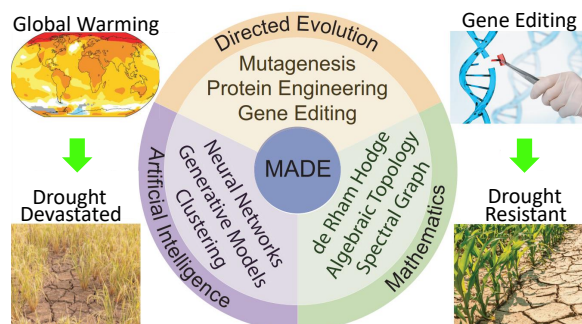


Figure 1: Illustration of mathematics-assisted directed evolution (MADE). Image source: (top left) https://bio.libretexts.org/Courses/Gettysburg_College/01%3A_Ecology_for_All/24%3A_Human_Global_Environment/24.02%3A_Implications_of_Climate_Change; (top right) <https://autism.org/funded-research-2010-2011/>; (bottom left) <https://www.shutterstock.com/search/drought-rice>; (bottom right) <https://news.mit.edu/2017/climate-change-drought-corn-yields-africa-0316>.

¹<https://sinews.siam.org/Details-Page/persistent-homology-analysis-of-biomolecular-data>

its incapability of coping with directed networks and digraphs (i.e., polarization, gene regulation), and its inability to characterize structured data (e.g., functional groups and protein domains and motifs). To address these challenges, many persistent topological Laplacians (PTLs) have been introduced, including persistent Laplacians [5], persistent path Laplacians, persistent sheaf Laplacians, persistent hypergraph Laplacians [6], persistent hyperdigraph Laplacians, and evolutionary de Rham-Hodge theory [7]. For example, persistent Laplacians, also known as persistent spectral graphs, can capture the homotopic shape evolution of data that cannot be described by PH (see Figure 2), giving rising to the accurate forecasting of the SARS-CoV-2 future dominant variants ².

The effectiveness and practicality of new TDA tools have been demonstrated in AIDE and AI-assisted protein engineering (AIPE) research [8]. Specifically, element-specific PH and element-specific persistent Laplacians were constructed to simplify molecular geometric complexes, reduce protein dimensionality, and capture topological invariants, shape evolution, and sequence disparities in the protein fitness landscape. These structure embeddings were com-

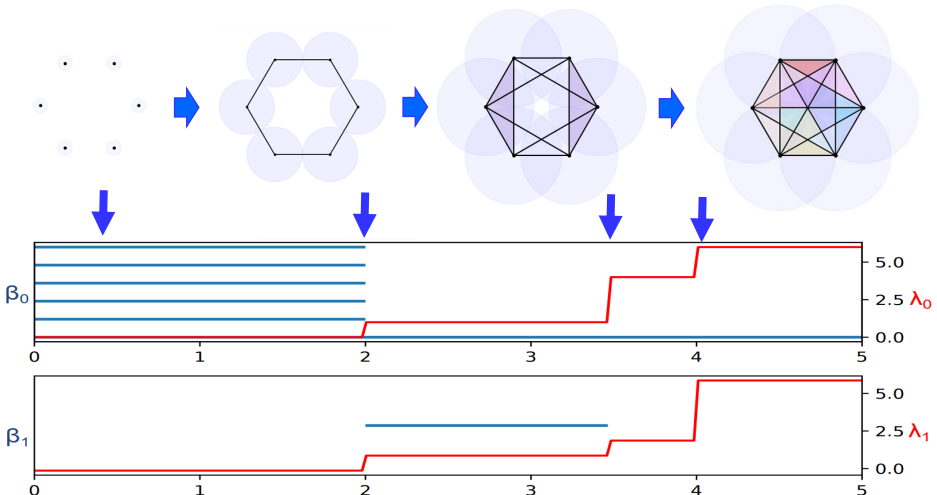


Figure 2: Comparison of persistent homology (PH) [3, 4] and persistent Laplacians (PLs) [5] for six points. The filtration characterized by the horizontal axis r of six points is shown in the top chart. The corresponding topological features of dimension 0 and dimension 1 are shown the second and third charts, respectively. PH barcodes ($\beta_0(r)$ and $\beta_1(r)$) are given in blue. The first non-zero eigenvalues of dimension 0 ($\lambda_0(r)$) and dimension 1 ($\lambda_1(r)$) of PLs are depicted in red. The harmonic spectra of PLs return all the topological invariants of PH, whereas the non-harmonic spectra of PLs capture the additional homotopic shape evolution of PLs during the filtration that are neglected by PH. Figure courtesy of Gengzhuo Liu.

combined with sequence embeddings obtained from advanced natural language processing (NLP) tools, such as autoencoders, long short-term memory, and transformers [2]. These methods were further integrated with cutting-edge machine learning algorithms to establish a new generation of AIDE and AIPE approaches [8].

Scientists are enthusiastic and optimistic about mathematics-assisted directed evolution and protein engineering [9]. Mathematicians can also play a crucial and distinctive role in this burgeoning field through the development of statistical models and/or mathematical frameworks for AI and machine learning. These models and frameworks are designed to optimize the efficiency and accuracy of machine learning algorithms [10]. Moreover, mathematicians can help capture the behavior of AI and machine learning systems and analyze their computational complexity, thereby providing invaluable insights into their capabilities and limitations. The continuous advances in mathematical techniques and innovative AI algorithms will determine the future of directed evolution and protein engineering and address some of the grand challenges of our time, such as climate change, food security, and pandemic prevention.

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